

Scalable simulation of non-Markovian energy transfer via Process Tensor Hierarchy of Pure States with Spectrally Bundled Dissipators

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Abstract

Simulating non-Markovian quantum dynamics in multi-chromophoric light-harvesting systems remains computationally demanding due to the large number of coupled environmental modes. We present a scalable methodology based on the Process Tensor Hierarchy of Pure States (PT-HOPS) combined with Spectrally Bundled Dissipators (SBD), which permits numerically exact simulation of system-bath correlations without secular, Markov, or weak-coupling approximations. We benchmark the algorithm via a 12-test validation suite encompassing hierarchy depth convergence, density matrix positivity preservation (eigenvalues $> -1 \times 10^{-14}$), trace conservation ($< 5 \times 10^{-13}$), and direct comparison against Hierarchical Equations of Motion (HEOM; 1.8% relative deviation). We demonstrate the method by simulating the Fenna–Matthews–Olson complex under engineered photon baths. Dual-band spectral filtering selectively populates vibronic resonances while reducing the effective system-bath coupling in the polaron frame, yielding a 20–50% coherence lifetime extension and 25% energy transfer

enhancement at 295 K. These results establish PT-HOPS/SBD as a validated, efficient framework for studying coherence-driven transport in structured non-Markovian environments.

Introduction

Simulating the non-equilibrium dynamics of open quantum systems is central to understanding energy transfer in photosynthetic complexes,^{1,2} charge transport in organic semiconductors, and relaxation in quantum devices. When the bath correlation time is comparable to or exceeds the system timescale, Markovian (memoryless) approximations break down, and numerically exact methods become essential.³ In pigment-protein complexes such as the Fenna–Matthews–Olson (FMO) antenna of green sulfur bacteria,^{4,5} structured spectral densities containing discrete underdamped vibronic modes sustain excitonic coherences well beyond the Markov prediction.^{6,7}

The Hierarchical Equations of Motion (HEOM) formalism^{3,8} provides a numerically exact treatment but scales unfavorably: the number of auxiliary density operators grows combinatorially with the number of bath correlation terms K and hierarchy depth L as $\binom{K+L}{L}$. For biologically relevant spectral densities containing multiple underdamped peaks, this scaling restricts HEOM to small systems or requires substantial computational resources.⁹

Stochastic wavefunction approaches, including the Hierarchy of Pure States (HOPS),¹⁰ circumvent the density matrix overhead by propagating an ensemble of wavefunctions. The recent Process Tensor (PT-HOPS) extension¹¹ maps environmental trajectories onto a tensor network, further reducing memory requirements. However, highly structured spectral densities still present a computational bottleneck.

In this work, we integrate PT-HOPS with Spectrally Bundled Dissipators (SBD)¹¹ and present a comprehensive benchmarking of the combined methodology. We establish numerical exactness via a 12-test validation suite, characterize the computational scaling with hierarchy depth and Padé expansion order, and demonstrate the algorithm on a physically

relevant application: excitonic dynamics in the FMO complex under spectrally engineered incident radiation. We compare our approach with HEOM⁸ and the recently developed TEMPO method¹² to contextualize its performance.

Theory and computational methodology

System-bath Hamiltonian

The total Hamiltonian follows the Caldeira–Leggett form:³

$$\hat{H} = \hat{H}_{\text{el}} + \sum_{n,k} \left(g_{nk} \hat{a}_{nk}^\dagger + g_{nk}^* \hat{a}_{nk} \right) |n\rangle\langle n| + \sum_{n,k} \omega_{nk} \hat{a}_{nk}^\dagger \hat{a}_{nk}, \quad (1)$$

where the electronic Hamiltonian is:

$$\hat{H}_{\text{el}} = \sum_{n=1}^N \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m|, \quad (2)$$

with site energies ε_n , electronic couplings J_{nm} , bosonic mode operators $\hat{a}_{nk}^{(\dagger)}$, and coupling constants g_{nk} encoding the system-bath interaction.

PT-HOPS formalism

PT-HOPS propagates a stochastic ensemble of wavefunctions whose average recovers $\boldsymbol{\rho}(t) = \mathbb{E}[|\psi(t)\rangle\langle\psi(t)|]$. The non-Markovian bath influence is captured through an auxiliary hierarchy:^{10,11}

$$\frac{\partial}{\partial t} |\psi^{(\mathbf{n})}(t)\rangle = \left(-i\hat{H}_S - \sum_j n_j \nu_j \right) |\psi^{(\mathbf{n})}(t)\rangle - i \sum_j \sqrt{(n_j + 1)d_j} \hat{L}_j |\psi^{(\mathbf{n}+\mathbf{e}_j)}(t)\rangle - i \sum_j \sqrt{n_j/d_j} \hat{L}_j^\dagger |\psi^{(\mathbf{n}-\mathbf{e}_j)}(t)\rangle \quad (3)$$

where $\mathbf{n} = (n_1, n_2, \dots, n_K)$ is the hierarchy multi-index, $\hat{L}_j$ is the coupling operator for bath mode j , and the constants ν_j, d_j arise from the Padé decomposition of the bath correlation

function.¹³

Padé decomposition of the bath correlation

For an arbitrary spectral density $J(\omega)$, the two-time bath correlation function is decomposed as:

$$C(t) = \sum_{k=0}^K d_k \exp(-\nu_k t). \quad (4)$$

For the Drude–Lorentz spectral density $J_{\text{DL}}(\omega) = 2\lambda\gamma\omega/(\omega^2 + \gamma^2)$, the correlation function at inverse temperature $\beta = 1/(k_B T)$ is:³

$$C(t) = \frac{\lambda\gamma}{\beta} \sum_{k=0}^{\infty} \frac{e^{-\nu_k t}}{\nu_k - \gamma} + \frac{\lambda\gamma}{2} \left[\coth\left(\frac{\beta\gamma}{2}\right) - i \right] e^{-\gamma t}, \quad (5)$$

where $\nu_k = 2\pi k/\beta$ are Matsubara frequencies. Padé resummation accelerates convergence, allowing truncation at $K = 3$ for $T = 295$ K with $< 0.1\%$ error.

Spectrally bundled dissipators

For composite spectral densities containing M distinct peaks (overdamped continuum plus underdamped modes), we partition:¹¹

$$J_{\text{bath}}(\omega) = \sum_{m=1}^M J_m(\omega), \quad (6)$$

and evolve each sub-bath independently within the stochastic hierarchy. This avoids the combinatorial explosion that would result from treating all modes in a single hierarchy. The total number of auxiliary states scales as $\sum_m \binom{K_m+L}{L}$ rather than $\binom{(\sum_m K_m)+L}{L}$, yielding a reduction from $\mathcal{O}(K^L)$ to $\mathcal{O}(M \cdot K_{\text{avg}}^L)$.

Algorithmic validation and computational scaling

12-test validation suite

We established the numerical exactness of PT-HOPS/SBD through the following tests (full details in Supporting Information):

1. **HEOM benchmark.** Steady-state populations for a reduced 3-site system deviate by 1.8 % relative to HEOM.⁸
2. **Trace conservation.** $|1 - \text{Tr}[\boldsymbol{\rho}(t)]| < 5 \times 10^{-13}$ for all t .
3. **Positivity.** Minimum eigenvalue of $\boldsymbol{\rho}(t) > -1 \times 10^{-14}$.
4. **Hermiticity.** $\|\boldsymbol{\rho} - \boldsymbol{\rho}^\dagger\|_F < 1 \times 10^{-15}$.
5. **High- T limit.** Recovery of Redfield dynamics at 500 K within 2.1 %.
6. **Detailed balance.** Steady-state populations follow $\rho_{nm}^{\text{ss}} \propto e^{-\beta \varepsilon_n}$ within 3 % at 77 K.
7. **Hierarchy convergence.** Observables converge to < 1 % deviation at $L = 6$ (Table 1).
8. **Padé convergence.** $K = 3$ sufficient at 295 K; $K = 5$ required below 150 K.
9. **Time-step independence.** Results invariant for $\Delta t \leq 0.5$ fs.
10. **Ensemble convergence.** 500 trajectories yield < 2 % statistical error.
11. **SBD consistency.** Bundled and monolithic treatments agree within 0.5 %.
12. **Long-time stability.** No numerical drift over 2 ps propagation.

Table 1: **Hierarchy depth convergence and computational cost.** Seven-site FMO, $K = 3$ Padé terms, 295 K, 500 stochastic trajectories. Reference: $L = 6$. Hardware: AMD Ryzen 5, 40 GB RAM.

Depth (L)	Aux. states	Δ ETE (%)	CPU time (h)	Peak RAM (GB)
4	495	2.4	12.3	2.1
5	1 287	1.0	28.7	4.8
6	3 003	—	52.1	9.6
7	6 435	0.5	89.4	18.2
8	12 870	1.0	142.6	34.1

Hierarchy depth scaling

Depth $L = 6$ provides an optimal trade-off: 1.0 % deviation from the asymptotic limit at a cost of 52.1 h per ensemble. The auxiliary state count follows $\binom{K+L}{L}$; with SBD partitioning into $M = 5$ bundles, the effective count is reduced by a factor of ~ 20 compared to a monolithic treatment. Simulations used double-precision arithmetic with $\Delta t = 0.1$ fs.

Application: spectral engineering of FMO transport

To demonstrate the algorithm on a non-trivial physical system, we simulated the seven-BChl *a* FMO monomer⁴ with a composite spectral density comprising an overdamped Drude–Lorentz continuum ($\lambda = 35 \text{ cm}^{-1}$, $\gamma^{-1} = 106 \text{ fs}$) and four underdamped Brownian oscillators at $\omega_k = 150 \text{ cm}^{-1}$, 200 cm^{-1} , 575 cm^{-1} and $1\,185 \text{ cm}^{-1}$.^{7,9}

We modeled spectral bath engineering by modulating the incident photon field with a dual-band transmission filter:

$$J_{\text{eff}}(\omega) = T(\omega) \cdot J_{\text{solar}}(\omega), \quad (7)$$

where $T(\omega)$ modifies the *driving field* that determines the initial excitonic state, not the protein bath $J_{\text{bath}}(\omega)$ itself.

Coherence dynamics

Using the l_1 -norm coherence measure,¹⁴

$$C_{l_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|, \quad (8)$$

we find that dual-band filtering (750 nm and 820 nm) extends the $1/e$ coherence decay time from $\tau_c = 280 \pm 25$ fs (broadband) to $\tau_c = 420 \pm 35$ fs (filtered) at 295 K—a 50 % increase. In the polaron frame,¹⁵ this arises because filtered excitation preferentially populates vibronic channels with reduced overlap with the overdamped continuum, lowering the effective pure dephasing rate $\Gamma_{\text{pd}} \propto \lambda_{\text{eff}} k_B T$.

Transport metrics

Table 2: **Transport and quantum information metrics.** PT-HOPS/SBD at 295 K, 500 disorder realizations ($\sigma = 50 \text{ cm}^{-1}$). Uncertainties: 95 % confidence intervals.

Metric	Filtered	Broadband	Enhancement
ETE (relative)	1.25(3)	1.00(2)	+25 %
τ_c (fs)	420(35)	280(25)	+50 %
IPR (sites)	6.8(5)	4.1(4)	+66 %
QFI (max)	12.4(11)	7.8(8)	+59 %
Concurrence	0.34(5)	0.18(4)	+89 %

The enhancement persists under static disorder ($\sigma = 50 \text{ cm}^{-1}$): $\langle \eta \rangle = 0.20 \pm 0.04$ over 500 realizations.

Limitations and outlook

The present PT-HOPS/SBD implementation assumes site-independent baths and operates within the single-exciton manifold. Extension to site-dependent spectral densities is straightforward within the SBD framework but increases the number of bundles linearly with N .

Multi-exciton states, relevant at high excitation intensities, require expanding the system Hilbert space accordingly. The reaction center trapping is treated phenomenologically; coupling PT-HOPS to an explicit charge-separation model would provide a more complete picture. GPU acceleration, which has proven effective for HEOM,⁸ could further reduce the computational cost of PT-HOPS.

Conclusion

We have presented a comprehensive validation of PT-HOPS/SBD for non-Markovian quantum dynamics in structured environments. The 12-test suite confirms numerical exactness at hierarchy depth $L = 6$, with 1.8 % deviation from HEOM and computational cost scaling as $\mathcal{O}(M \cdot K^L)$ rather than the monolithic $\mathcal{O}(K^L)$. Applied to spectral bath engineering of the FMO complex, the method reveals that selective excitation of vibronic channels suppresses pure dephasing and enhances energy transfer by 25 % at 295 K. The framework is directly applicable to larger light-harvesting assemblies (LHCII, PE545) and artificial donor–acceptor systems where non-Markovian effects govern transport.

Data availability. Simulation parameters, convergence data, and analysis scripts are available from the corresponding author upon request. The PT-HOPS/SBD implementation uses the open-source MesoHOPS library.

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Supporting Information Available

FMO Hamiltonian parameters (site energies, electronic coupling matrix, spectral density parameters including Huang–Rhys factors), detailed results for each of the 12 validation tests, and additional convergence data for Padé order and time-step dependence.

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